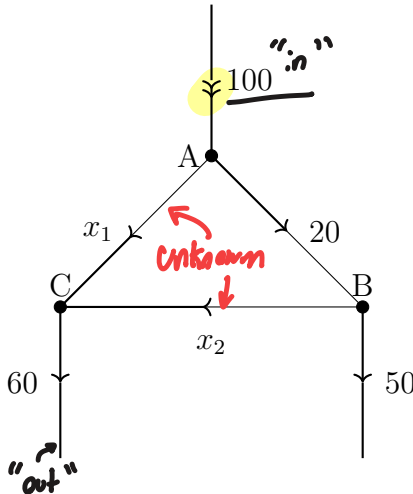




Solution to linear system does not always exist:



flow rate conservation
at each node

$$A : 100 = 20 + x_1$$

$$B : 20 = x_2 + 50$$

$$C : x_1 + x_2 = 60$$

↙ solve
for x

$$\text{node A : } 100 = x_1 + 20 \quad \rightarrow \quad x_1 = 80$$

$$\text{node B : } 20 = x_2 + 50 \quad \rightarrow \quad x_2 = -30$$

$$\text{node C : } x_1 + x_2 = 60 \quad \rightarrow \quad 80 + (-30) = +50 \neq 60$$

doesn't work

Reason:

$\therefore$ the system isn't
consistent

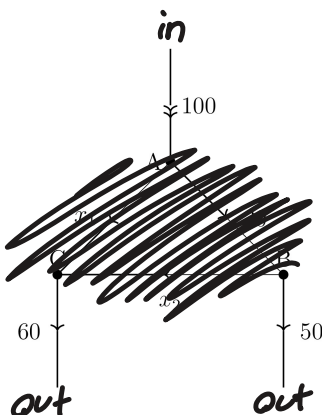
"IN"

100

$\neq$

"OUT"

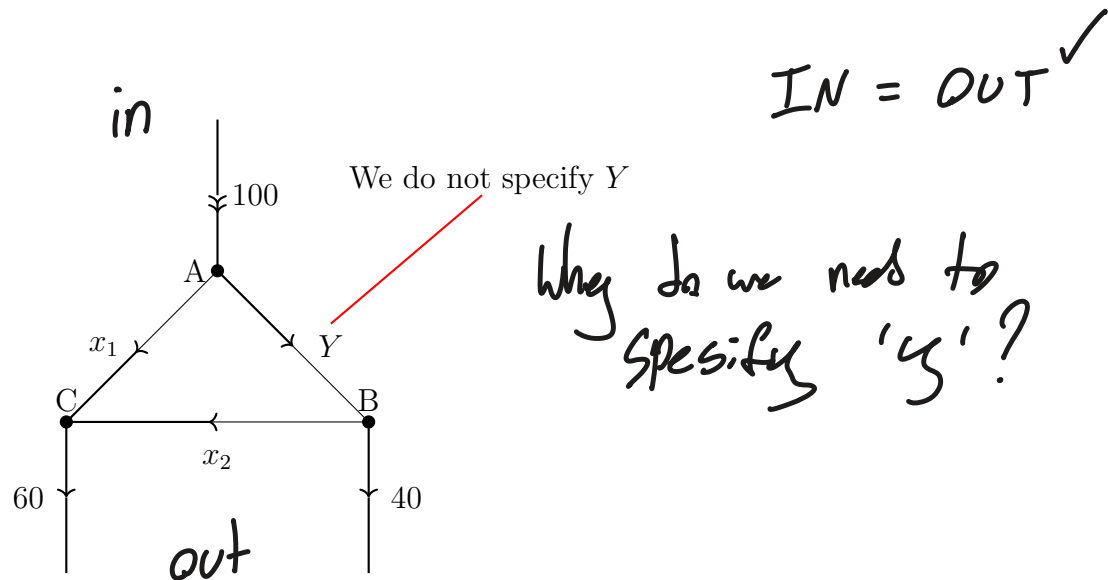
$60 + 50 = 110$



$$100 \neq 50 + 60$$

we can tell it's not
consistent very easy

Solution to linear system is not always unique:



$$\text{node A : } 100 = x_1 + Y \rightarrow x_1 = 100 - Y$$

$$\text{node B : } Y = x_2 + 40 \rightarrow x_2 = Y - 40$$

$$\text{node C : } x_1 + x_2 = (100 - Y) + (Y - 40) = 60 \rightarrow \checkmark \cong 60$$

$\Rightarrow$ True for any Y

Consistent?

Pipe size will change
flow rate, $\therefore$ you must state 'Y'

"IN"

"OUT"

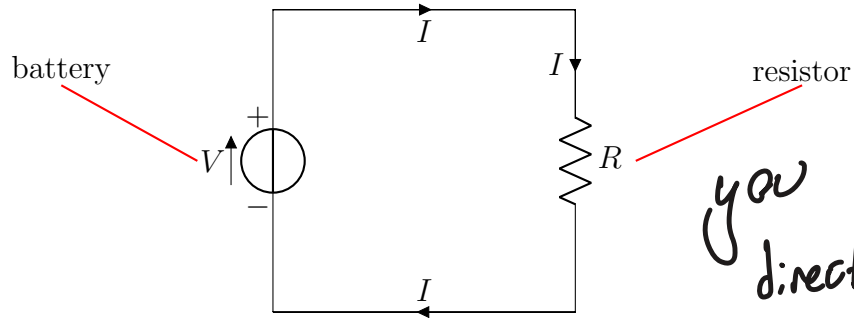
$$100 = 60 + 40 = 100$$

$\Rightarrow$ In the course we study when the system of linear equations is consistent, and when there is a unique solution

$\Rightarrow$ In previous example setting $Y = 20$ gives a unique solution; without specifying Y system of equations is underdetermined

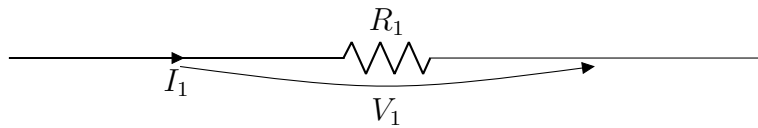
Electric circuits

⇒ Elementary electric circuit: *current*



Amps

- **current** I flows from the positive pole of the battery to the negative pole
- battery is characterized by the **voltage** V (electric potential difference between the poles)
- current is characterized by a flow of positively charged particles (convention) — negatively charged particles (electrons) move in the direction opposite to that of the current
- **resistor** R reduces the *electric potential* according to **Ohm's law**:



$$\begin{array}{ccccc}
 \textcircled{V_1} & = & \textcircled{R_1} & \cdot & I_1 \\
 \text{potential drop } V_1 & & \text{resistance of the resistor } R_1 & & \text{current } I_1 \\
 \text{across the resistor} & & \text{property of the material} & & \text{through the resistor} \\
 \text{in volts (V)} & & \text{in Ohms } (\Omega) & & \text{in amperes (A)}
 \end{array}$$

Along the wire
current cannot change

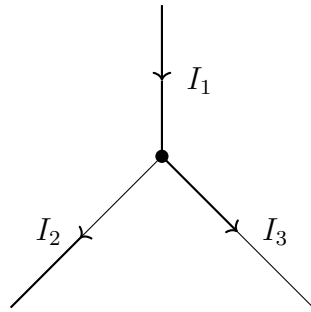
7

review stuff
ez cake

Kirchoff's laws of electric circuits:

① Current conservation at each node

$$I_{\text{IN}} = I_{\text{OUT}}$$

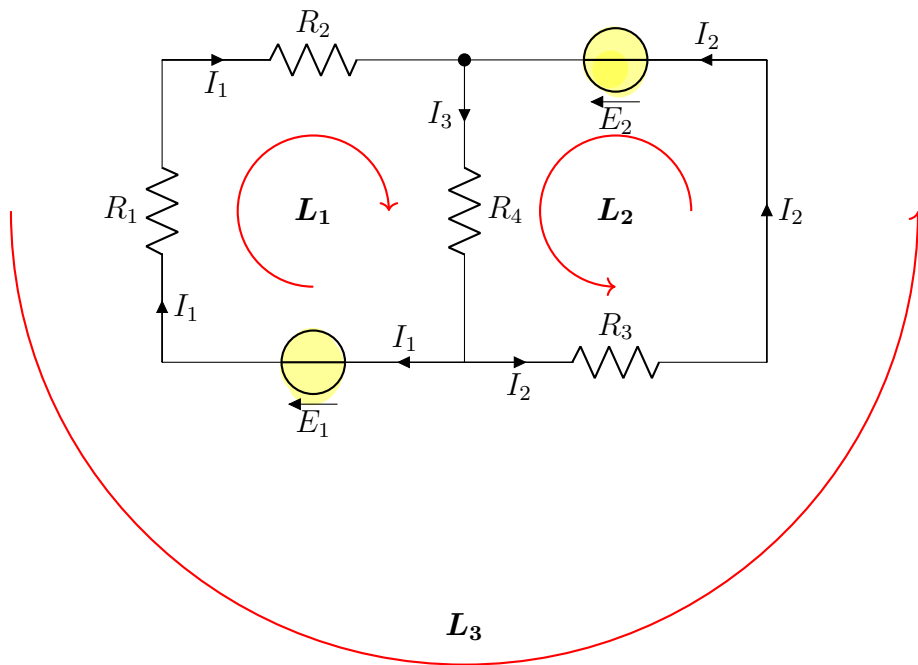


Current can only change at junction

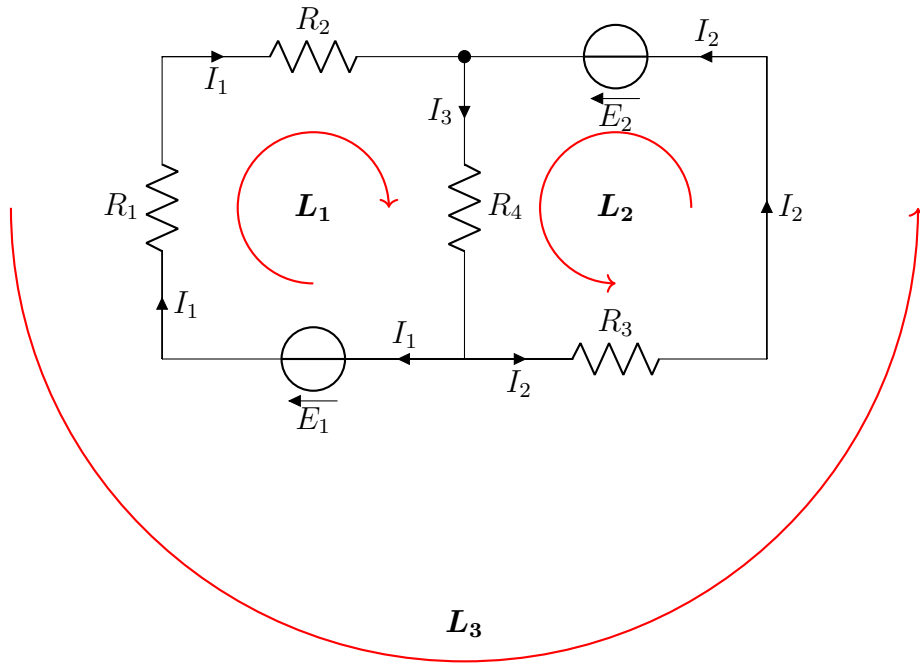
$$\underbrace{I_1}_{\text{"in"}} = \underbrace{I_2 + I_3}_{\text{"out"}}$$

② Sum of voltage rises = sum of voltage drops

$\Rightarrow$ In normal language: electric potential change is zero for any closed loop of the circuit



$\Rightarrow$ There are 3 closed loops: $\{L_1, L_2, L_3\}$; orientation in red



$$\mathbf{L_1 :} \quad \underbrace{R_1 \cdot I_1 + R_2 \cdot I_1 + R_4 \cdot I_3}_{\text{"voltage drops"}} = \underbrace{E_1}_{\text{"voltage rises"}}$$

$$\mathbf{L_2 :} \quad \underbrace{R_4 \cdot I_3 + R_3 \cdot I_2}_{\text{"voltage drops"}} = \underbrace{E_2}_{\text{"voltage rises"}}$$

$$\mathbf{L_3 :} \quad \underbrace{R_3 \cdot I_2 + (-R_2 \cdot I_1) + (-R_1 \cdot I_1)}_{\text{"voltage drops"}} = \underbrace{(-E_1) + E_2}_{\text{"voltage rises"}}$$

$\Rightarrow$ Note **minus** signs: they appear because we are going 'against' the current direction

$\Rightarrow$ Note:

$$\mathbf{L_3 = L_2 - L_1}$$

that is, loop $\mathbf{L_3}$ is not *independent*